ST 625 Project Survival Patterns in Diabetes Patients

Lai On Ping

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Introduction

Diabetes is a long-term metabolic disorder due to prolong high blood glucose level over time. It is a very common disease and cause of mortality. Most diabetes patients relied on insulin to maintain healthy. Studying the patterns of diabetes patients is important for optimizing treatment strategies and informing public health policies. This study is to estimate and compare the survival probabilities between people whom received insulin treatments and not using both parametric and non-parametric tests. Furthermore, I would like to estimate the hazard ratios and their confidence intervals for different covariates to assess whether certain covariates can explain the survival probabilities.

The dataset from the National Institute of Diabetes and Digestive and Kidney Diseases has diagnostic measurements of 768 diabetes patients. All patients are females at least 21 years old of Pima Indian heritage.

%%Smith, J.W., Everhart, J.E., Dickson, W.C., Knowler, W.C., & Johannes, R.S. (1988). Using the ADAP learning algorithm to forecast the onset of diabetes mellitus. In Proceedings of the Symposium on Computer Applications and Medical Care (pp. 261–265). IEEE Computer Society Press.

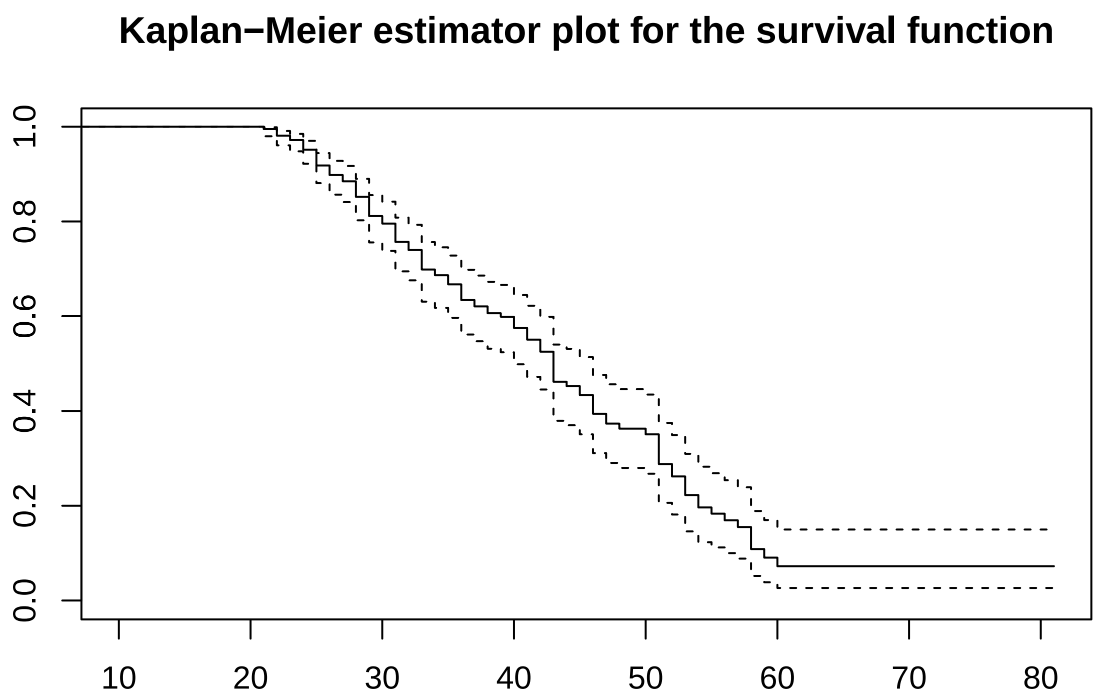
Table 1: Dataset "diabetes"

Variables	**Information**
Pregnancies	Number of pregnancies
Glucose	2 hr Plasma glucose concentration (mg/dL)
BloodPressure	diastolic blood pressure (mmHg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2 hr serum insulin
BMI	Body mass index (kg/m^2)
DiabetesPedigreeFunction	diabetes likelihood depends on the subject's age and family history
Age	this would be "time" in the data analysis
Outcome	two levels factor with alive or death.

The dataset has a crucial problem. The "variable" Insulin has a well known relationship with diabetes. The higher Insulin measurement a person has, the stronger insulin resistance, the more severer diabetes problem. However, there are about half of the insulin values are zero in this dataset, which should be considered as missing data. (Glucose and blood pressure has similar issue but there were not too many missing values.) It leads to a dilemma that whether variable deletion or dropping missing data is more appropriate. Therefore, there are a few steps for this project. First, use the complete data to fit different models and test the significance of "Insulin". After that, choose an appropriate model to perform residual analysis and make inference.

Significance of Insulin

After removing missing data, there are 392 observations. To give insight of the data, the Kaplan-Meier estimator plot for the survival function for the complete dataset has the following:



Below is a summary result table of different models with Forward and Backward Direction. Forward Direction compares with an intercept model and a model with "Insulin"; Backward Direction compares with full model and a model without Insulin.

Table 2: Result of LRT test significance of Insulin

	Forward	**Backward**
Weibull Reg	$\times$	$*$
AFT	$\times$	$\times$
Cox ph	$\times$	$*$

$\times$: no evidence ; $*$: little evidence

Weibull Regression models the Weibull hazard function and assume the effect of covariates on the hazard rate is constant over time (Proportional hazards assumption). The harzard of Weibull distribution is

$$\lambda(t) = \alpha\beta t^{\alpha-1}$$

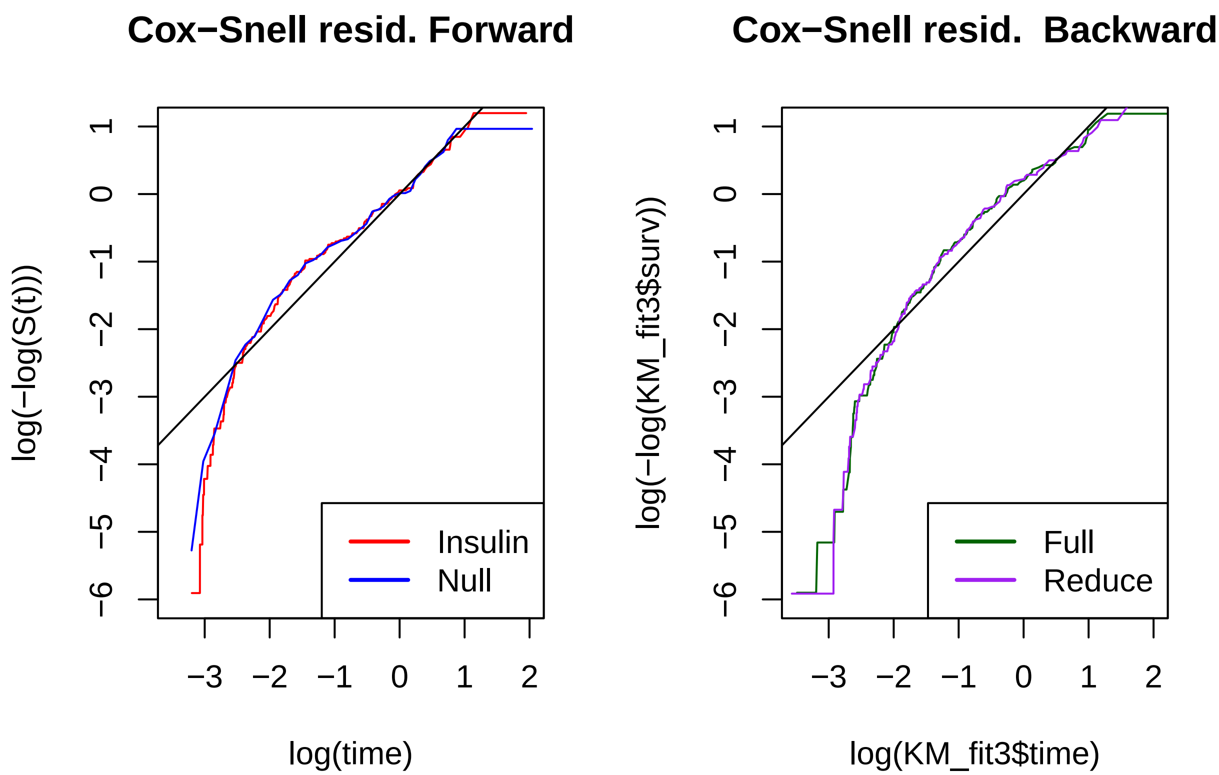
where alpha is the shape parameter, and beta is the rate parameter.

log-logistic AFT model estimates the time scale/acceleration when event occurs. It assumed that covariates are multiplicative and constant over time and residuals follow exponential(1).

$$\log(X_i) = \mu + z_i^T\eta + \sigma\epsilon$$

Cox proportional hazards models the hazard ratio and assumes hazard ratio is constant overtime.

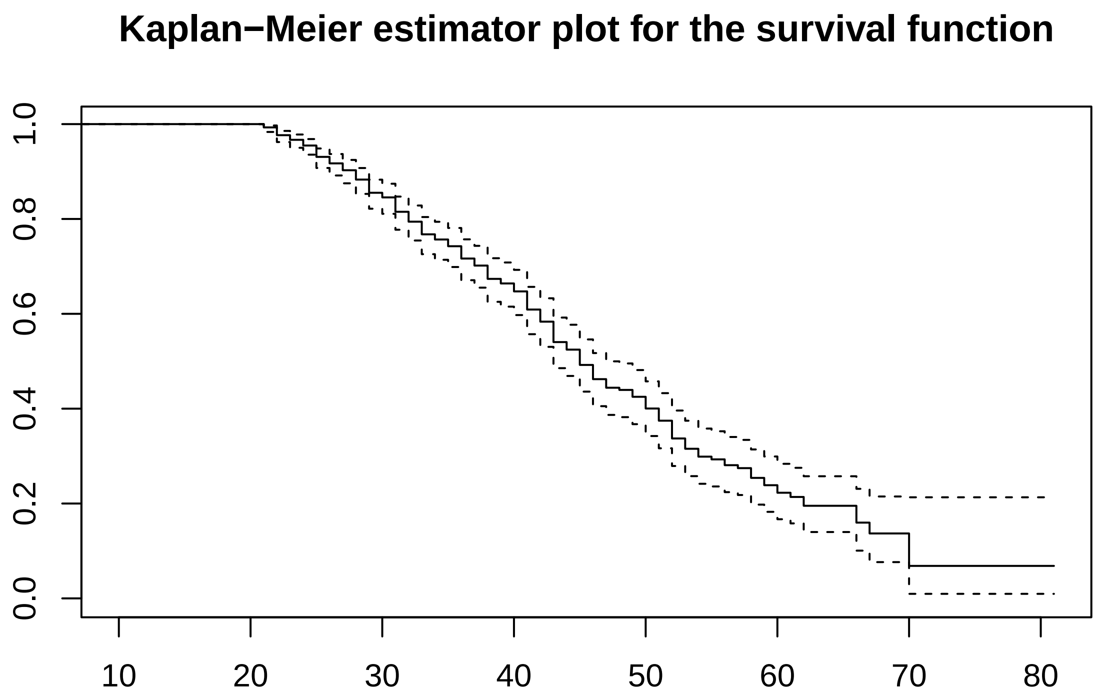
The following is cox-snell plot for Weibull Regression models:



Based on the plot, there is not much difference between the models with Insulin and without Insulin, but there is difference between full model and null model, indicates that other variables may have effects on the survival time. The discussion will save for later. At his point, it is appropriate to drop the variable "Insulin" based on 392 observations. However, note that backward directions both suggest a little evidence of violation of proportional hazard assumption.

With All data

By dropping "Insulin", there are 724 observations to fit the models. This sections would explore the best models to capture the patterns of survival times of diabetes patient. As an insight, the Kaplan Meier Estimator plot is the following:



Comparing to the plot from last section, there are better estimate of survival probabilities, especially between age 60 to 70.

Weibull Regression

Maximum Likelihood Estimator

If the diabetes death follows Weibull distribution with shape parameter alpha and rate parameter beta, then

```
## The MLE for alpha is: 3.599329  with SE 0.1540383
```

```
## The MLE for lambda is: 6.591658e-07 with SE 3.93577e-07
```

If we would like to test the hypothesis alpha = 1, both Likelihood Ratio test and Wald's test suggest strong evidence that the shape parameter is not 1.

Regression in R

```r
Wei_full <- survreg(Surv(Age, Outcome)~.-Insulin, data = diabetes2)
```

The output suggest that Pregnancies, Glucose, Blood Pressure, and BMI have significance effect on the hazard. The significance also explains why the SK plot from earlier section would look different between null model and a complex model. Therefore, refitting the model would obtain:

```
##
## Call:
## survreg(formula = Surv(Age, Outcome) ~ . - Insulin - SkinThickness -
##     DiabetesPedigreeFunction, data = diabetes2)
##                   Value Std. Error      z       p
## (Intercept)    4.402788   0.135192  32.57 < 2e-16
## Pregnancies    0.012828   0.004704   2.73  0.0064
## Glucose       -0.002232   0.000523  -4.27 1.9e-05
## BloodPressure  0.006796   0.001383   4.91 8.9e-07
## BMI           -0.021991   0.002357  -9.33 < 2e-16
## Log(scale)    -1.370512   0.043532 -31.48 < 2e-16
##
## Scale= 0.254
##
## Weibull distribution
## Loglik(model)= -1098   Loglik(intercept only)= -1159.1
##  Chisq= 122.17 on 4 degrees of freedom, p= 1.8e-25
## Number of Newton-Raphson Iterations: 6
## n= 724
```

The regression coefficient for BMI is -0.021991, indicates a decrease in risk of death at a scale of 0.978249 for one-unit increase of BMI while other covariates remain the same.

Cox ph Model

Fitting the model

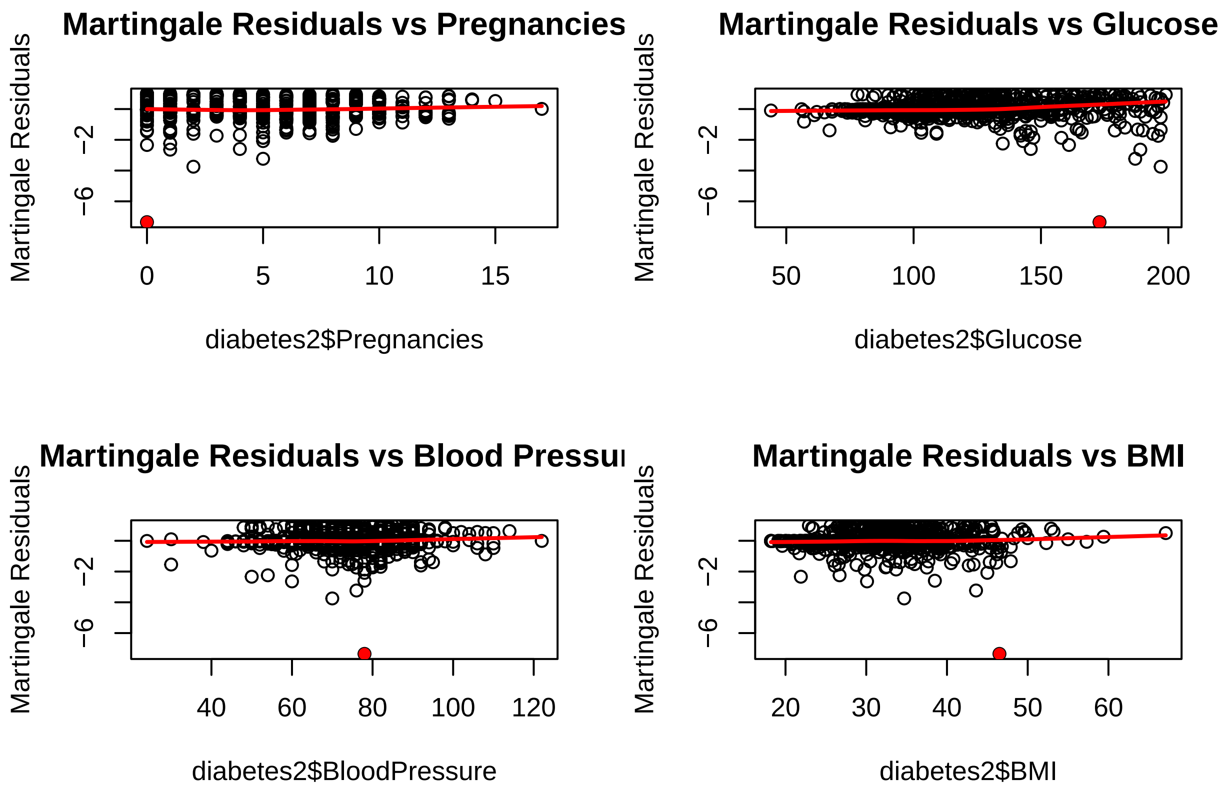
By fitting the all covariates (except Insulin), R suggests SkinThickness and DiabetesPedigreeFunction are not significance. Refitting would obtain:

```
## Call:
## coxph(formula = Surv(Age, Outcome) ~ . - Insulin - SkinThickness -
##     DiabetesPedigreeFunction, data = diabetes2)
##
##                    coef exp(coef)  se(coef)      z        p
## Pregnancies   -0.062279  0.939621  0.018951 -3.286  0.00102
## Glucose        0.009702  1.009749  0.001947  4.983 6.27e-07
## BloodPressure -0.024931  0.975377  0.005525 -4.513 6.40e-06
## BMI            0.078102  1.081233  0.009679  8.069 7.08e-16
##
## Likelihood ratio test=115.9  on 4 df, p=< 2.2e-16
## n= 724, number of events= 249
```

The regression coefficient for BMI is 0.078102, indicates an increase in hazard at a scale of 1.081233 for one-unit increase of BMI while other covariates remain the same.

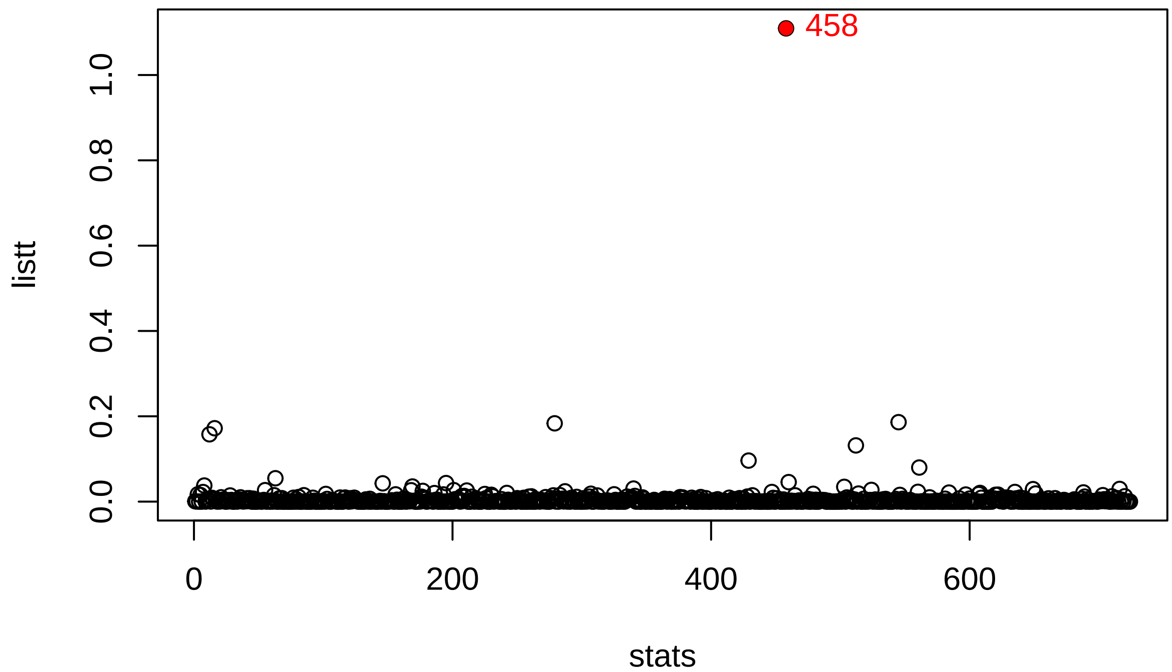
However, what is the goodness of fit of this model?

Martingale residual



There are no significant patterns for Pregnancy, Blood Pressure, nor BMI, but Glucose seemed to have a non-linear increasing trend. Moreover, all the plots suggest there is an outlier. This is observation 458.

The influence function agrees with the martingale residual:



Removing the outlier and refit the model did have some effects on the numerical values of the estimate, but did not change the directions or conclusion of the covariates.

Discussion

Both Coxph model and Weibull Regression model suggests Pregnancies, Glucose , Blood Pressure, and BMI have effect on the survival time, while there is no evidence of significance of Insulin, Skin Thickness , and Diabetes Pedigree Function. It is reasonable for Skin Thickness does not effecting survival time, since it is also a result/symptom of diabetes. Diabetes Pedigree Function also makes sense since it aimed to predict probability of diabetes, but all patients in this dataset have already been diagnosed diabetes. Insulin is complicated. In the first section, Insulin itself has not evidence on affecting survival time in the forward direction, but found a little evidence in backward direction. A plausible reason is that Insulin have relationships with other covarites. For example, Insulin would decrease glucose level, but patients would have Insulin resistance, so the body was not able to digest glucose in the body and therefore has high glucose and Insulin level at the same time, which is literaly the symptom of diabetes. Therefore, Insulin itself may not have effect on survival, but do has effect on other measurements that would affect survival time, which creates multicollinearity problem.

Due the length constraint, the AFT model was not included in the second major section.

Appendix

Section 1

```r
#Remove NA data, which were written as 0 in the data.
NAs <- c(which(diabetes$Glucose == 0),
         which(diabetes$BloodPressure == 0),
         which(diabetes$BMI == 0),
         which(diabetes$Insulin == 0))
NAs <- unique(NAs)
diabetes1 <- diabetes[-NAs,]

plot(survfit(Surv(Age, Outcome)~1,
             data = diabetes1,conf.type = "log-log"),
     xlim=c(10,81),
     main= "Kaplan-Meier estimator plot for the survival function")

###LRT
#Forward WRM
WRMIns <- survreg(Surv(Age, Outcome)~Insulin, data = diabetes1)
G <- 2*(WRMIns$loglik[2]- WRMIns$loglik[1])
p_val <- 1- pchisq(G,df=1)
#In Forward direction, given test statistic 0.85 with degree of freedom 1 and p-value 0.36, there is no

#Backwar WRM
WRMfull <- survreg(Surv(Age, Outcome)~., data = diabetes1)
WRMreduce <- survreg(Surv(Age, Outcome)~. - Insulin, data = diabetes1)
G <- 2*(WRMfull$loglik[2]- WRMreduce$loglik[2])
p_val <- 1- pchisq(G,df=1)
#In Backward direction, given test statistic 2.83 with degree of freedom 1 and p-value 0.09249506, ther


#Forward AFT
AFTIns <- survreg(Surv(Age, Outcome)~Insulin, data = diabetes1, dist = "loglogistic")
G <- 2*(AFTIns$loglik[2]- AFTIns$loglik[1])
p_val <- 1- pchisq(G,df=1)
#In Forward direction, given test statistic 0.99 with degree of freedom 1 and p-value 0.32, there is no

#Backward AFT
AFTfull <- survreg(Surv(Age, Outcome)~., data = diabetes1, dist = "loglogistic")
AFTred <- survreg(Surv(Age, Outcome)~. -Insulin, data = diabetes1, dist = "loglogistic")
G <- 2*(AFTfull$loglik[2]- AFTred$loglik[2])
p_val <- 1- pchisq(G,df=1)
#In Backward direction, given test statistic 1.177196 with degree of freedom 1 and p-value 0.28, there

#Coxph
cox_full <- coxph(Surv(Age, Outcome)~., data = diabetes1)
cox_red <- coxph(Surv(Age, Outcome)~.- Insulin, data = diabetes1)
G <- 2*(cox_full$loglik[2]- cox_red$loglik[2])
p_val <- 1- pchisq(G,df=1)
#cox_null <- coxph(Surv(Age, Outcome)~1, data = diabetes1)
cox_Ins <- coxph(Surv(Age, Outcome)~Insulin, data = diabetes1)
#A forward direction has p-value 0.5775, suggests no evidence that Insulin is significant to the propor
```

```r
#A backward direction has p-value 0.07, suggests a little evidence that Insulin is significant to the p


#SK plots
#Insulin
tilde_t_i <- -log(pweibull(t_i, 1 / WRMIns$scale, scale = predict(WRMIns), lower.tail = FALSE))
KM_fit1 <- survfit(Surv(tilde_t_i, delta_i)~1)

#Null
WRMnull <- survreg(Surv(Age, Outcome)~1, data = diabetes1)
tilde_t_i <- -log(pweibull(t_i, 1 / WRMnull$scale, scale = predict(WRMnull), lower.tail = FALSE))
KM_fit2 <- survfit(Surv(tilde_t_i, delta_i)~1)

#Full
tilde_t_i <- -log(pweibull(t_i, 1 / WRMfull$scale, scale = predict(WRMfull), lower.tail = FALSE))
KM_fit3 <- survfit(Surv(tilde_t_i, delta_i)~1)

#reduce
tilde_t_i <- -log(pweibull(t_i, 1 / WRMreduce$scale, scale = predict(WRMreduce), lower.tail = FALSE))
KM_fit4 <- survfit(Surv(tilde_t_i, delta_i)~1)

#Forward plot
par(mfrow=c(1,2))
plot(log(KM_fit1$time),log(-log(KM_fit1$surv)),type="l",ylab = "log(-log(S(t)))", xlab = "log(time)",
main = "Cox-Snell resid. Forward",xlim=c(-3.5,2),ylim=c(-6,1), col="red")
lines(log(KM_fit2$time),log(-log(KM_fit2$surv)),type="l", col="blue")
abline(0,1,col="black")
legend("bottomright", legend = c("Insulin", "Null"), col = c("red", "blue"), lwd = 2)

#Backward plot
plot(log(KM_fit3$time),log(-log(KM_fit3$surv)),type="l", col = "darkgreen",
main = "Cox-Snell resid.  Backward",xlim=c(-3.5,2),ylim=c(-6,1))
lines(log(KM_fit4$time),log(-log(KM_fit4$surv)),type="l", col = "purple")
abline(0,1,col="black")
legend("bottomright", legend = c("Full", "Reduce"), col = c("darkgreen", "purple"), lwd = 2)
```

```r
NAss <- c(which(diabetes$Glucose == 0),
          which(diabetes$BloodPressure == 0),
          which(diabetes$BMI == 0))
NAss <- unique(NAss)
diabetes2 <- diabetes[-NAss,]


plot(survfit(Surv(Age, Outcome)~1,
             data = diabetes2,conf.type = "log-log"),
     xlim=c(10,81),
     main= "Kaplan-Meier estimator plot for the survival function")

#survreg's scale = 1/(rweibull shape)
#survreg's intercept = log(rweibull scale)

Weib0 <-  survreg(Surv(Age, Outcome)~1, data = diabetes2)
alpha_hat <- 1/Weib0$scale
```

```
lambda_hat <- sum(diabetes2$Outcome)/sum(diabetes2$Age^alpha_hat)

ti <- diabetes2$Age
d <- diabetes2$Outcome
row1 <- c(sum(d)/alpha_hat^2 + lambda_hat*sum((log(ti)^2)*ti^alpha_hat), sum(log(ti)*ti^alpha_hat))
row2 <- c(sum(log(ti)*ti^alpha_hat), sum(d)/lambda_hat^2)
FI <- rbind(row1,row2)
inverse <- solve(FI)

cat("The MLE for alpha is:",alpha_hat," with SE", sqrt(inverse[1,1]) ,"\n")
cat("The MLE for lambda is:",lambda_hat,"with SE" , sqrt(inverse[2,2]), "\n")

#LRT
exp0 <- survreg(Surv(Age, Outcome)~1, data = diabetes2, dist="exponential")
G <- 2*(Weib0$loglik[1] - exp0$loglik[1])
1-pchisq(G, df = 1)

#Wald
W <- (alpha_hat -1)^2/sqrt(inverse[1,1])
1-pchisq(W, df = 1)

Wei_full <- survreg(Surv(Age, Outcome)~.-Insulin, data = diabetes2)

Wei_red <- survreg(Surv(Age, Outcome)~. -Insulin -SkinThickness -DiabetesPedigreeFunction , data = diabe
summary(Wei_red)

Cox <- coxph(Surv(Age, Outcome)~. -Insulin , data = diabetes2)
Cox

COX <-  coxph(Surv(Age, Outcome)~. -Insulin - SkinThickness - DiabetesPedigreeFunction, data = diabetes2
COX


martin <- residuals(COX, type = "martingale")
outliers <- which(martin < -6)

par(mfrow=c(2,2))
plot(x=diabetes2$Pregnancies, y=martin, ylab = "Martingale Residuals",
     main = "Martingale Residuals vs Pregnancies")
lines(lowess(diabetes2$Pregnancies, martin), col = "red", lwd=2)
points(x= diabetes2$Pregnancies[outliers], y = martin[outliers], col ="red", pch=16)

plot(x=diabetes2$Glucose, y=martin, ylab = "Martingale Residuals",
     main = "Martingale Residuals vs Glucose")
lines(lowess(diabetes2$Glucose, martin), col = "red", lwd=2)
points(x= diabetes2$Glucose[outliers], y = martin[outliers], col ="red", pch=16)


plot(x=diabetes2$BloodPressure, y=martin, ylab = "Martingale Residuals",
     main = "Martingale Residuals vs Blood Pressure")
lines(lowess(diabetes2$BloodPressure, martin), col = "red", lwd=2)
points(x= diabetes2$BloodPressure[outliers], y = martin[outliers], col ="red", pch=16)
```

```
plot(x=diabetes2$BMI, y=martin, ylab = "Martingale Residuals",
     main = "Martingale Residuals vs BMI")
lines(lowess(diabetes2$BMI, martin), col = "red", lwd=2)
points(x= diabetes2$BMI[outliers], y = martin[outliers], col ="red", pch=16)


beta_resid <- resid(COX, type = "dfbeta")
info <- solve(vcov(COX))
holder <- beta_resid%*%info%*%t(beta_resid)
listt <- diag(holder)
outliers <- which(listt > 1)
plot(listt, xlab = "stats")
points(outliers,listt[outliers], col = "red",pch=16)
text(outliers, listt[outliers], labels = outliers, pos = 4, col = "red")


diabetes3 <- diabetes2[-outliers,]
COX2 <-  coxph(Surv(Age, Outcome)~. -Insulin - SkinThickness - DiabetesPedigreeFunction, data = diabetes
COX
COX2
```